

# Data Driven Turbulence Modeling for Shear Mixing

Internship Project Report

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## Abstract

Understanding the effects of turbulence is critical in turbomachinery applications, particularly for predicting flow behavior across fan and blade rows. Reynolds-Averaged Navier–Stokes (RANS) solvers are the standard because of their low computational cost. But they are limited in their ability to accurately model anisotropic turbulent stresses. High-fidelity Large-Eddy Simulation (LES) and Direct Numerical Simulation (DNS) resolve these structures but are much more computationally expensive.

This study develops a machine learning framework to recover anisotropic stresses from RANS solver outputs. The model is trained and validated using an experimental dataset from NASA and UIUC’s turbulent shear mixing experiment. Corresponding RANS simulations of the experimental cases provide the input features, while the experimental measurements serve as validation targets.

Results show that the model can reconstruct anisotropic turbulence structures beyond the plane-of-strain limitation inherent to RANS solvers. The model offers a pathway to weave shear mixing closures into RANS solutions for compressible shear flows relevant to turbomachinery. This allows us to model trailing-edge environments without incurring the computational cost of LES or DNS.

# Nomenclature

## Notation Conventions:

- $\overline{(\cdot)}$  Time-averaged quantity
- $(\cdot)'$  Fluctuating component (Reynolds decomposition)
- $\widetilde{(\cdot)}$  Favre-averaged (density-weighted) quantity
- $(\cdot)_{\text{ref}}$  Reference value (temperature, velocity, length, etc.)

| Symbol        | Definition                         |
|---------------|------------------------------------|
| $u$           | Streamwise velocity component      |
| $v$           | Vertical velocity component        |
| $w$           | Spanwise velocity component        |
| $U$           | Velocity magnitude                 |
| $p$           | Pressure                           |
| $\rho$        | Density                            |
| $T$           | Temperature                        |
| $\mu$         | Dynamic viscosity                  |
| $\nu$         | Kinematic viscosity                |
| $S_{ij}$      | Strain-rate tensor                 |
| $\Omega_{ij}$ | Rotation-rate tensor               |
| $R_{ij}$      | Reynolds stress tensor             |
| $a_{ij}$      | Anisotropy tensor                  |
| $b_{ij}$      | Normalized anisotropy tensor       |
| $k$           | Turbulent kinetic energy           |
| $x, y, z$     | Cartesian coordinates              |
| $M$           | Mach number                        |
| $M_c$         | Convective Mach number             |
| $\text{Re}$   | Reynolds number                    |
| $\dot{m}$     | Mass flow rate                     |
| $a$           | Speed of sound                     |
| $\tau_{ij}$   | Viscous diffusion tensor           |
| $\mu_t$       | Turbulent viscosity                |
| $\gamma$      | Ratio of specific heats            |
| $R$           | Specific gas constant              |
| $c_p$         | Specific heat at constant pressure |
| $\lambda_i$   | Eigenvalues of anisotropy tensor   |
| $II_a, III_a$ | Invariants of $a_{ij}$             |
| $II_b, III_b$ | Invariants of $b_{ij}$             |

| <b>Acronym</b> | <b>Meaning</b>                                 |
|----------------|------------------------------------------------|
| DNS            | Direct Numerical Simulation                    |
| FARANS         | Favre-Averaged Reynolds-Averaged Navier–Stokes |
| LDV            | Laser Doppler Velocimetry                      |
| LES            | Large Eddy Simulation                          |
| NASA           | National Aeronautics and Space Administration  |
| RANS           | Reynolds-Averaged Navier–Stokes                |
| UIUC           | University of Illinois at Urbana–Champaign     |
| FOV            | Field of View                                  |

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# 1 Introduction

In turbomachinery, a major source of loss occurs at the trailing edge of turbine blades, where pressure-side and suction-side flows merge and form turbulent shear mixing layers [1]. Understanding the turbulent shape and structure of these layers is crucial for improving efficiencies.

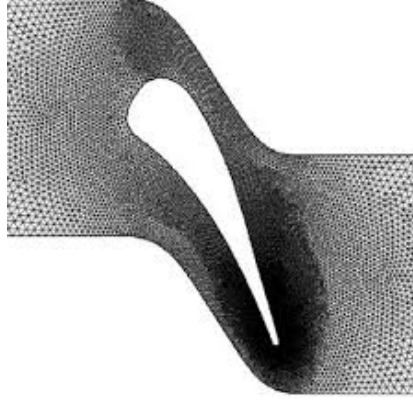


Figure 1: NASA C3X turbine blade used in related aerodynamic and heat transfer experiments.

RANS solvers are the workhorse of turbine blade and cascade design due to their low computational cost [7]. However, their closure in most turbulence models relies on the Boussinesq hypothesis, which relates Reynolds stress anisotropy directly to mean strain rate [2]. This assumption constrains the predicted structure of turbulence onto the plane of strain in the Lumley triangle [3]. With most turbulence models like  $k-\omega$  SST or  $k-\epsilon$ , the Boussinesq hypothesis prevents RANS from capturing the full range of anisotropic structures observed in shear mixing layers [8].

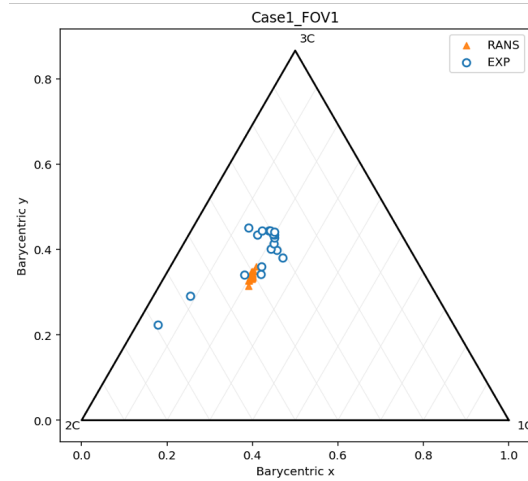


Figure 2: Lumley triangle representation of turbulence states. Traditional RANS closures (Boussinesq hypothesis) restrict predictions to the plane of strain, while our objective is to recover realizable turbulence states beyond this limitation.

Our approach is to develop a data-driven closure of RANS turbulence modeling from



experimental data using machine learning. By learning experimental Reynolds stress anisotropy directly from RANS solver outputs, a machine learning model will have the ability to produce anisotropy that is not constrained by the plane-of-strain. The goal of this work is to be applied into analysis of turbine blade trailing edge analysis.

## 2 NASA, UIUC Compressible Mixing Layer Experiments

### 2.1 Experiment Setup

The NASA, UIUC compressible mixing layer experiments were performed in a wind tunnel designed to study how two air streams at different Mach numbers mix. A plate separated the high and low-speed streams as they developed to testing conditions. The pressures of each stream were matched at the inlets to avoid unwanted pressure gradients. The speed difference between these two streams is described by the convective Mach number:

$$M_c = \frac{U_1 - U_2}{a_1 + a_2} \quad (1)$$

which represents the difference in speeds normalized by the average speed of sound in the two streams.

Velocity data was collected using Laser Doppler Velocimetry (LDV), which measures particle speeds in the flow. This provided velocity components in the streamwise and vertical directions, capturing mean flow characteristics such as  $\bar{u}$ ,  $\bar{v}$ , and turbulence statistics including  $\overline{u'u'}$ ,  $\overline{v'v'}$ , and  $\overline{u'v'}$ . Measurements were taken at multiple fields of view along the test section to capture how the shear layer grew and evolved downstream.

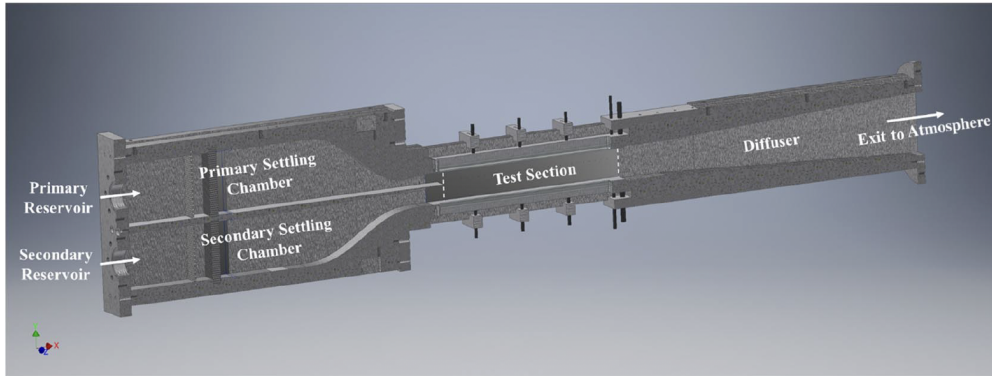


Figure 3: Schematic of the NASA/UIUC compressible mixing layer experimental setup, showing the wind tunnel layout, splitter plate, and measurement regions.

### 2.2 Data Assumptions and Implications

It is important to note that the experimental dataset, while corresponding to mildly compressible flow, reports regular Reynolds stresses rather than density-weighted (Favre) stresses. This assumption simplifies the modeling, as it neglects significant density fluctuations that would be present in higher convective Mach number cases.

Table 3: **Compressible mixing layer Boundary conditions and uncertainties.**

| Input Parameters    | Case 1            | Case 2            | Case 3            | Case 4            | Case 5            |
|---------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| $M_1$               | $0.463 \pm 0.012$ | $1.003 \pm 0.021$ | $1.571 \pm 0.025$ | $1.955 \pm 0.021$ | $2.463 \pm 0.032$ |
| $M_2$               | $0.089 \pm 0.009$ | $0.189 \pm 0.009$ | $0.285 \pm 0.014$ | $0.269 \pm 0.008$ | $0.175 \pm 0.009$ |
| $P_{01}$ (kPa)      | $109.32 \pm 0.05$ | $151.84 \pm 0.05$ | $270.41 \pm 0.05$ | $445.50 \pm 0.06$ | $778.10 \pm 0.09$ |
| $P_{02}$ (kPa)      | $94.56 \pm 0.05$  | $82.47 \pm 0.05$  | $71.47 \pm 0.05$  | $63.83 \pm 0.05$  | $50.95 \pm 0.05$  |
| $P_1$ (kPa)         | $93.94 \pm 0.10$  | $80.37 \pm 0.11$  | $62.02 \pm 0.10$  | $57.58 \pm 0.10$  | $50.84 \pm 0.11$  |
| $P_2$ (kPa)         | $93.92 \pm 0.10$  | $80.59 \pm 0.11$  | $66.29 \pm 0.10$  | $61.51 \pm 0.10$  | $49.91 \pm 0.11$  |
| $T_{01}$ (K)        | $296.10 \pm 0.50$ | $294.60 \pm 0.50$ | $284.59 \pm 0.50$ | $298.02 \pm 0.50$ | $289.25 \pm 0.51$ |
| $T_{02}$ (K)        | $292.48 \pm 0.50$ | $293.58 \pm 0.50$ | $295.57 \pm 0.50$ | $298.22 \pm 0.50$ | $292.58 \pm 0.50$ |
| $T_1$ (K)           | $283.95 \pm 0.84$ | $245.25 \pm 2.04$ | $190.50 \pm 2.68$ | $168.87 \pm 2.26$ | $130.70 \pm 2.63$ |
| $T_2$ (K)           | $292.02 \pm 0.51$ | $291.49 \pm 0.54$ | $290.85 \pm 0.69$ | $293.96 \pm 0.56$ | $290.80 \pm 0.54$ |
| $U_1$ (m/s)         | $156.25 \pm 4.3$  | $314.89 \pm 6.32$ | $434.76 \pm 6.08$ | $509.24 \pm 4.35$ | $564.16 \pm 4.59$ |
| $U_2$ (m/s)         | $30.35 \pm 3.0$   | $64.75 \pm 2.96$  | $97.28 \pm 4.87$  | $92.57 \pm 2.79$  | $59.76 \pm 3.19$  |
| $r = U_2/U_1$       | $0.194 \pm 0.020$ | $0.206 \pm 0.010$ | $0.224 \pm 0.012$ | $0.182 \pm 0.006$ | $0.106 \pm 0.006$ |
| $s = \rho_2/\rho_1$ | $0.972 \pm 0.004$ | $0.844 \pm 0.007$ | $0.700 \pm 0.010$ | $0.614 \pm 0.008$ | $0.441 \pm 0.010$ |
| $M_c$               | $0.185 \pm 0.008$ | $0.381 \pm 0.011$ | $0.546 \pm 0.013$ | $0.690 \pm 0.009$ | $0.883 \pm 0.010$ |

Table 4: Boundary conditions for the NASA/UIUC compressible mixing layer experiments

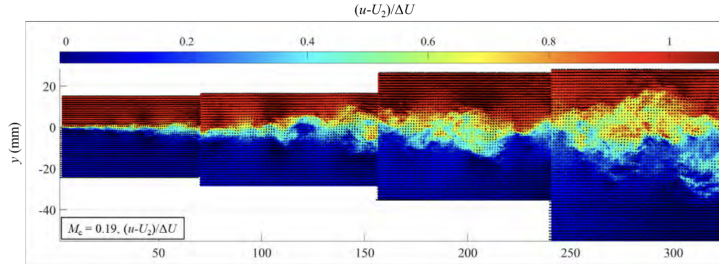


Figure 4: Normalized velocity fields for case  $M_c \approx 0.38$  in the NASA/UIUC mixing layer dataset

## 3 Methodology

### 3.1 RANS Simulations Matching the Experiment

To provide input features for the machine learning model, we ran RANS cases matched to the boundary conditions of the experimental configurations. A  $k$ - $\omega$  SST turbulence model was used, with geometry, inflow, and thermodynamic conditions aligned to the experimental setup (see Figure 3).

### 3.2 Assumptions

Air is modeled as an **Ideal Gas** for feature engineering purposes, with the constant specific gas constant  $R$ .

$$\bar{p} = \bar{\rho} R \bar{T}, \quad (2)$$

**Dynamic viscosity** depends on temperature via Sutherland’s hypothesis. This allows our model to work around the turbulent viscosity modeling.

$$\mu(T) = \mu_{\text{ref}} \left( \frac{T}{T_{\text{ref}}} \right)^{3/2} \frac{T_{\text{ref}} + S}{T + S}, \quad (3)$$

where  $T_{\text{ref}}$  and  $\mu_{\text{ref}}$  are reference temperature and viscosity, and  $S$  is Sutherland’s constant. For air, we take  $S \approx 110$  K.

**Density fluctuations.** In accordance with the NASA/UIUC experimental dataset, density fluctuations in the Reynolds stresses are treated as negligible. This assumption is consistent with the available data, which includes only the velocity correlations ( $uu$ ,  $uv$ ,  $vv$ , etc.), and allows the stresses to be modeled as  $\bar{\rho} \overline{u'_i u'_j}$  without explicit  $\rho'$  contributions.

### 3.3 Feature Engineering

Our data sources separate into two groups: (i) experimental measurements and (ii) RANS simulations of those experiments. The experimental dataset provides raw Reynolds stresses and higher-order turbulence quantities, from which we extract anisotropy tensors and Lumley triangle mappings (see Appendix A). For the RANS dataset, input features were derived from the steady Favre-averaged Navier–Stokes (FARANS) momentum equation,

$$\frac{\partial}{\partial x_j} (\bar{\rho} \tilde{u}_i \tilde{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} - \frac{\partial \overline{\rho u''_i u''_j}}{\partial x_j}. \quad (4)$$

The divergence of the mean momentum flux can be decomposed into advection and convection terms:

$$\frac{\partial}{\partial x_j} (\rho u_i u_j) = \rho u_j \frac{\partial u_i}{\partial x_j} + \rho u_i \frac{\partial u_j}{\partial x_j},$$

where the first term corresponds to **advection** (transport of  $u_i$  by  $u_j$ ) and the second to **convection**, associated with dilatation and compressibility effects.

From this balance we compute the following feature groups:

- **Advection:**  $\bar{\rho} \tilde{u}_j \partial \tilde{u}_i / \partial x_j$

- **Convection:**  $\bar{\rho} \tilde{u}_i \partial \tilde{u}_j / \partial x_j$
- **Pressure gradient:**  $\partial \bar{p} / \partial x_i$
- **Viscous diffusion:**  $\partial \bar{\tau}_{ij} / \partial x_j$
- **Turbulent transport (closure term):**  $\partial \overline{\rho u_i'' u_j''} / \partial x_j$
- **Derived tensors:** strain rate  $S_{ij}$ , rotation rate  $\Omega_{ij}$ , and invariants.

To improve consistency, gradients were recomputed with a Moving Least Squares (MLS) algorithm across the RANS grid, which reduced noise from overlapping control volumes.

Although RANS solvers directly output some gradients, overlapping control volumes can distort their distributions. To improve consistency, gradients were recomputed using a Moving Least Squares (MLS) algorithm across the RANS grid, yielding more robust derivative estimates (see Appendix C for details).

Within the momentum advection term, we expand using the velocity gradient identity  $\partial u_i / \partial x_j = S_{ij} + R_{ij}$  to separate strain and rotation contributions. This gives

$$\rho u_j \frac{\partial u_i}{\partial x_j} = \rho u_j S_{ij} + \rho u_j R_{ij},$$

where the first term represents **strain-induced advection** (stretching of turbulent structures), and the second term represents **rotation-induced advection** (vortical transport). In addition, we include the bulk **convection term**,  $\rho u_i \partial u_j / \partial x_j$ , which accounts for compressibility effects through dilatation.

## 3.4 Non-Dimensionalization and Normalization

### 3.4.1 Non-Dimensional Groups

To ensure consistent scaling, all fields were first non-dimensionalized using reference values derived from the inlet boundary conditions (Table x). The reference length scale was defined as the mass-flux averaged inlet diameter:

$$L_{\text{ref}} = \frac{\sum_i \dot{m}_i D_i}{\sum_i \dot{m}_i}, \quad (5)$$

where  $D_i$  is the local inlet diameter and  $\dot{m}_i$  is the corresponding mass flux.

For scalar fields such as velocity, temperature, and pressure, reference values were taken as mass-flux averages at the inlet:

$$\phi_{\text{ref}} = \frac{\sum_i \dot{m}_i \phi_i}{\sum_i \dot{m}_i}. \quad (6)$$

With these definitions, nondimensional forms are given by:

$$\hat{u}_i = \frac{u_i}{U_{\text{ref}}}, \quad (7)$$

$$\hat{T} = \frac{T}{T_{\text{ref}}}, \quad (8)$$

$$\hat{p} = \frac{p}{p_{\text{ref}}}, \quad (9)$$

$$\hat{x}_i = \frac{x_i}{L_{\text{ref}}}, \quad (10)$$

$$\widehat{\frac{\partial \phi}{\partial x_j}} = \frac{L_{\text{ref}}}{\phi_{\text{ref}}} \frac{\partial \phi}{\partial x_j}. \quad (11)$$

Here the derivative scaling follows directly from the chain rule, ensuring that gradients of nondimensional variables remain consistent with their corresponding scalar normalizations. To close, the steady nondimensional FARANS momentum equation is:

$$\frac{\partial}{\partial x_j} (\bar{\rho} \tilde{u}_i \tilde{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{1}{Re_0} \frac{\partial \tilde{\tau}_{ij}}{\partial x_j} - \frac{\partial}{\partial x_j} \left( \widetilde{\rho u_i'' u_j''} \right), \quad (12)$$

where  $Re_0 = \rho_{\text{ref}} U_{\text{ref}} L_{\text{ref}} / \mu_{\text{ref}}$  is the Reynolds number. This expression emphasizes how all terms in the governing equations are consistently scaled with the chosen reference values.

### 3.4.2 Normalization of Model Inputs

After non-dimensionalization, all features were globally normalized using dataset-wide extrema. This choice preserves the relative magnitudes between features (e.g., pressure vs. velocity), which is important for maintaining physically interpretable scaling across cases.

Since feature-engineered gradient terms often exhibit extreme outliers, a clipping method based on each feature's kurtosis was applied to define effective minima and maxima. This approach prevented the normalized dataset from collapsing too tightly into the  $[0, 1]$  range, while ensuring that the vast majority of non-outlier values still mapped within the interval. Further details and illustrative distributions are provided in Appendix B.

### 3.4.3 Model Inputs

Table 5 summarizes the final set of input features used for all modeling in this work. These were the features used for the trials denoted in table 6. All  $z$ -components were omitted due to the quasi-two-dimensional dataset geometry. Each feature was normalized using reference-based non-dimensionalization, as described in Sec. 3.4.

## 3.5 Machine Learning Framework

For this analysis we used a fully connected neural network (multi-layer perceptron). The architecture consisted of 15 layers with 10 neurons per layer. Models were trained with

Table 5: Final set of input features passed into the model.

| Expression                             | Name                       | Normalization     |
|----------------------------------------|----------------------------|-------------------|
| $\rho u_j S_{ij}$                      | Strain-induced advection   | Reference scaling |
| $\rho u_j R_{ij}$                      | Rotation-induced advection | Reference scaling |
| $\rho u_i \partial u_j / \partial x_j$ | Convection                 | Reference scaling |
| $\tau_{ij}$                            | Viscous stress tensor      | Reference scaling |
| $\partial \bar{p} / \partial x_i$      | Pressure gradient          | Reference scaling |
| $\bar{p}$                              | Static pressure            | Reference scaling |
| $\bar{u}_i$                            | Velocity components        | Reference scaling |
| $\mu(T)$                               | Sutherland viscosity       | Reference scaling |
| $\bar{\rho}$                           | Density                    | Reference scaling |

the Adam optimizer and a step-based learning rate scheduler, for a total of 6000 epochs. This framework also supports to Kolmogorov–Arnold Networks (KANs). However, with larger input sizes, the model displayed no significant improvement compared to the MLP.

### 3.6 Loss Formulation

Our learning objective is to recover realizable anisotropy from RANS data. To do so, we must produce not only accurate Reynolds stresses, but also stresses that follow the anisotropic structure described in Pope’s *Turbulent Flows*. Accordingly, the composite objective is organized into three groups: *data*, *physics*, and *constraints*. Data corresponds to direct comparison against experimental truth, physics terms are derived from model outputs, and constraints enforce realizability conditions such as zero trace and eigenvalue consistency within the Lumley triangle.

The total loss is defined as

$$\mathcal{L} = w_{\text{data}} \mathcal{L}_{\text{data}} + w_{\text{phys}} \mathcal{L}_{\text{phys}} + w_{\text{const}} \mathcal{L}_{\text{const}}, \quad (13)$$

with optional per-component and per-case weights. All group weights are manually initialized and static to provide a stable learning curriculum.

#### 3.6.1 Data Term

The model outputs turbulent kinetic energy ( $k$ ) and anisotropy ( $a_{ij}$ ). The data loss is given by

$$\mathcal{L}_{\text{data}} = \sum_{j \in \mathcal{C}} w_{a_{ij}} \mathcal{L}(a_{ij}) + w_k \mathcal{L}(k), \quad (14)$$

where  $\mathcal{C}$  indexes the six independent tensor components.

#### 3.6.2 Component Weights.

Weights for anisotropy components are initialized by inverse mean scaling with log smoothing:

$$w_{a_{ij}} = \log_b \left( \frac{\bar{a}_{ij}}{\max(\bar{a}) + \epsilon} \right). \quad (15)$$

For components with near-zero mean (e.g.  $a_{xz}$ ,  $a_{yz}$ ), the weight was statically set above unity (on the order of 1.5 in the smoothed scale) to ensure that these terms were not under-weighted. The turbulent kinetic energy weight  $w_k$  was kept larger than any  $w_{a_{ij}}$ , making TKE conservation the primary training priority.

### 3.6.3 Physics Terms

Physics losses were applied by reconstructing Reynolds stresses ( $R_{ij}$ ) and the normalized anisotropy tensor ( $b_{ij}$ ) from the model predictions of  $k$  and  $a_{ij}$ . These provide a pathway to enforce consistency between predicted quantities and physically derived tensors. When enabled, the losses are

$$\mathcal{L}_{\text{RST}} = \|\hat{R}_{ij} - R_{ij}\|, \quad (16)$$

$$\mathcal{L}_{b_{ij}} = \|\hat{b}_{ij} - b_{ij}\|, \quad (17)$$

where  $\hat{R}_{ij}$  and  $\hat{b}_{ij}$  are reconstructed from  $k_{\text{pred}}$  and  $a_{ij}^{\text{pred}}$ .

### 3.6.4 Constraint Terms

Constraint losses were designed to enforce realizability conditions and maintain the correct mathematical structure of turbulence anisotropy. These include:

$$\mathcal{L}_{\text{trace}} = \|\text{tr}(\hat{a}_{ij})\|, \quad (18)$$

$$\mathcal{L}_{\text{inv } a} = \|\hat{I}_2^a - I_2^a\| + \|\hat{I}_3^a - I_3^a\|, \quad (19)$$

$$\mathcal{L}_{\text{inv } b} = \|\hat{I}_2^b - I_2^b\| + \|\hat{I}_3^b - I_3^b\|, \quad (20)$$

$$\mathcal{L}_{\text{eig } a} = \|\hat{\lambda}(a) - \lambda(a)\|, \quad (21)$$

$$\mathcal{L}_{\text{eig } b} = \|\hat{\lambda}(b) - \lambda(b)\|, \quad (22)$$

where  $I_2$  and  $I_3$  denote the second and third invariants. Here,  $\mathcal{L}_{\text{inv}}$  terms use invariants computed directly from the tensor components, while  $\mathcal{L}_{\text{eig}}$  terms use invariants reconstructed from the eigenvalues. This separation allows the model to be penalized both on component-level realizability and on the spectral properties of anisotropy.

### 3.6.5 Case Weights.

In addition to component and group weights, case weights are applied across all training points depending on which case they belong to (e.g. Case 1, Case 2). These weights are designed so that no case is treated as less important simply because it has fewer points or lower overall turbulence intensity. The formulation is

$$w_c = \frac{\overline{k_c}}{\max(\overline{k_c}) + \epsilon}, \quad (23)$$

with  $\epsilon = e^{-12}$  to avoid division by zero.

This normalization ensures that the optimizer views each case equally, forcing the model to learn relationships that generalize across cases rather than placing importance on larger magnitude scales or denser sampling.

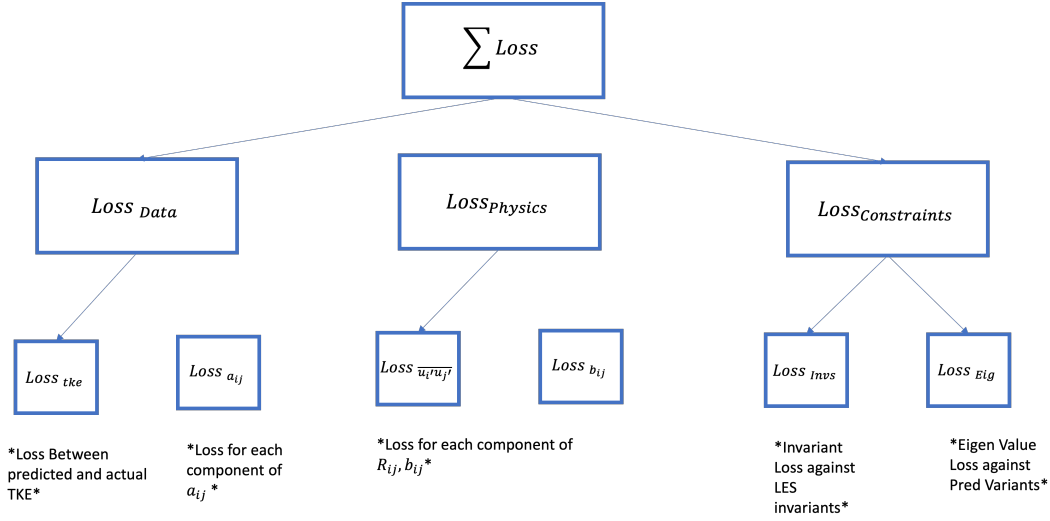


Figure 5: Loss hierarchy: data  $\rightarrow$  physics  $\rightarrow$  constraints.

### 3.7 Loss Hierarchy

## 4 Trials and Testing Strategy

The experimental dataset was collected field-of-view (FOV) by FOV. We therefore designed training and testing trials around the ability to interpolate and extrapolate depending on which FOVs are included in the training set.

In single-case trials, one case is used for training while individual FOVs from that case are held out for testing. If the held-out FOV is interior (e.g. FOV 2 or 3), the test represents interpolation; if the held-out FOV is at the boundary (FOV 1 or 4), the test represents extrapolation.

Multi-case trials expand this idea by training on all FOVs from one case and a subset of FOVs from another. The model is then tested on the remaining FOVs of the second case, which can again represent either interpolation or extrapolation depending on which FOVs are excluded.

All trials were performed using the set of input features summarized in Table 5, with  $z$ -components omitted as described in Section 3.4.3.

Table 6: Summary of training and testing trials. Interpolation corresponds to holding out interior FOVs (2 or 3), while extrapolation corresponds to holding out boundary FOVs (1 or 4).

| Trial ID             | Training Set        | Testing Set   | Type    |
|----------------------|---------------------|---------------|---------|
| Single-Case-Interp-1 | Case 2: FOVs 1,3,4  | Case 2: FOV 2 | Interp. |
| Single-Case-Extrap-1 | Case 2: FOVs 1,2,3, | Case 2: FOV 4 | Extrap. |



## 5 Results

### 5.1 Single-Case Interpolation Results

In this trial, interior FOVs were held out, requiring the model to interpolate within the spatial domain of a single case. This is expected to be the more favorable test since the training data bounds the held-out region.

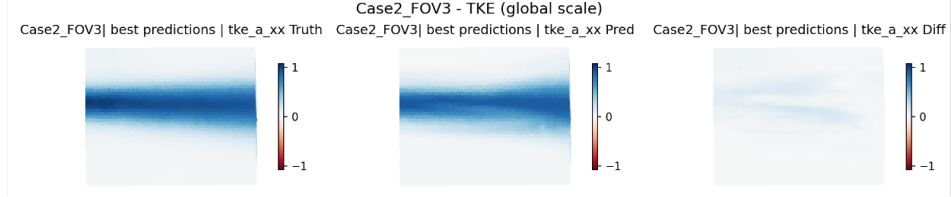


Figure 6: Predicted versus true turbulent kinetic energy (TKE) for the interpolation case.

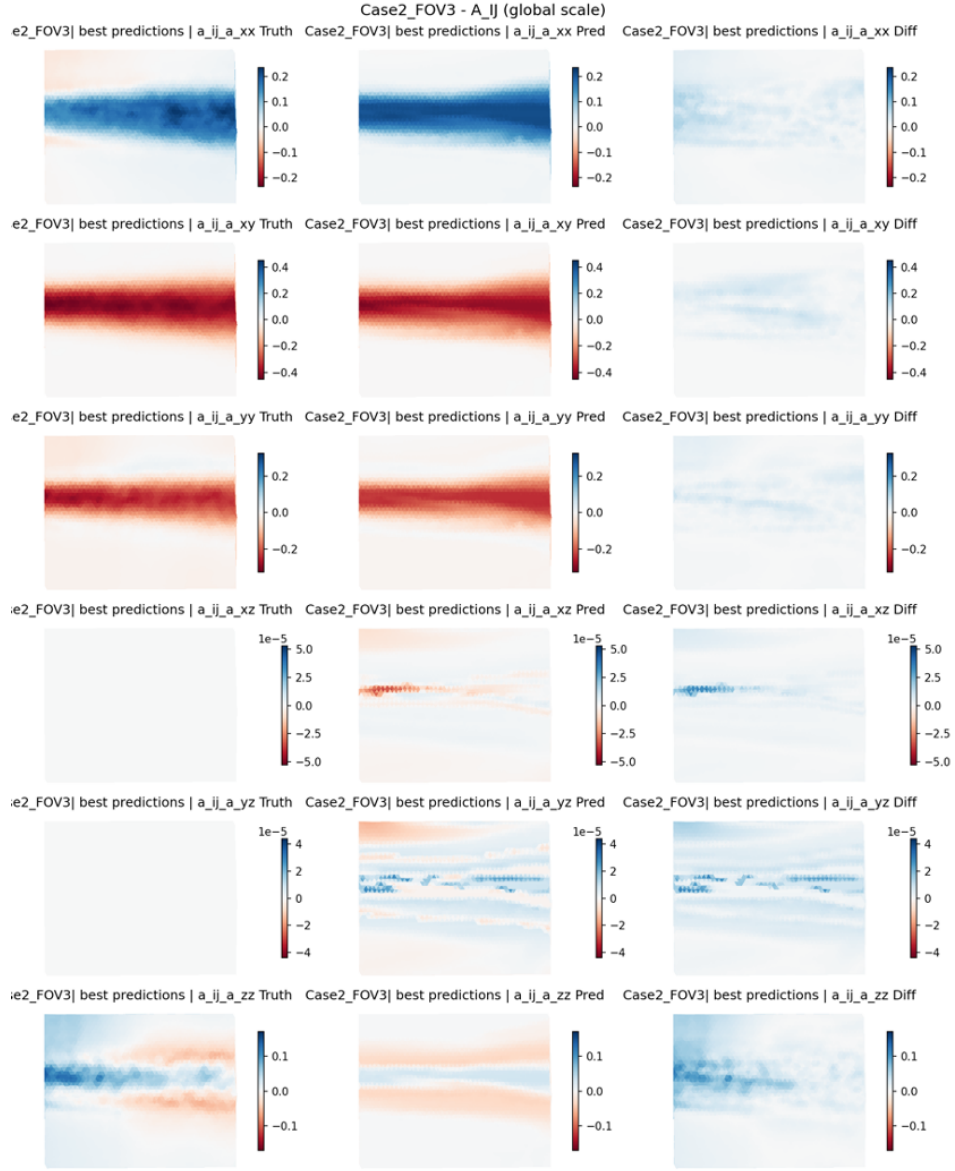


Figure 7: Predicted versus true anisotropy for the interpolation testing case.

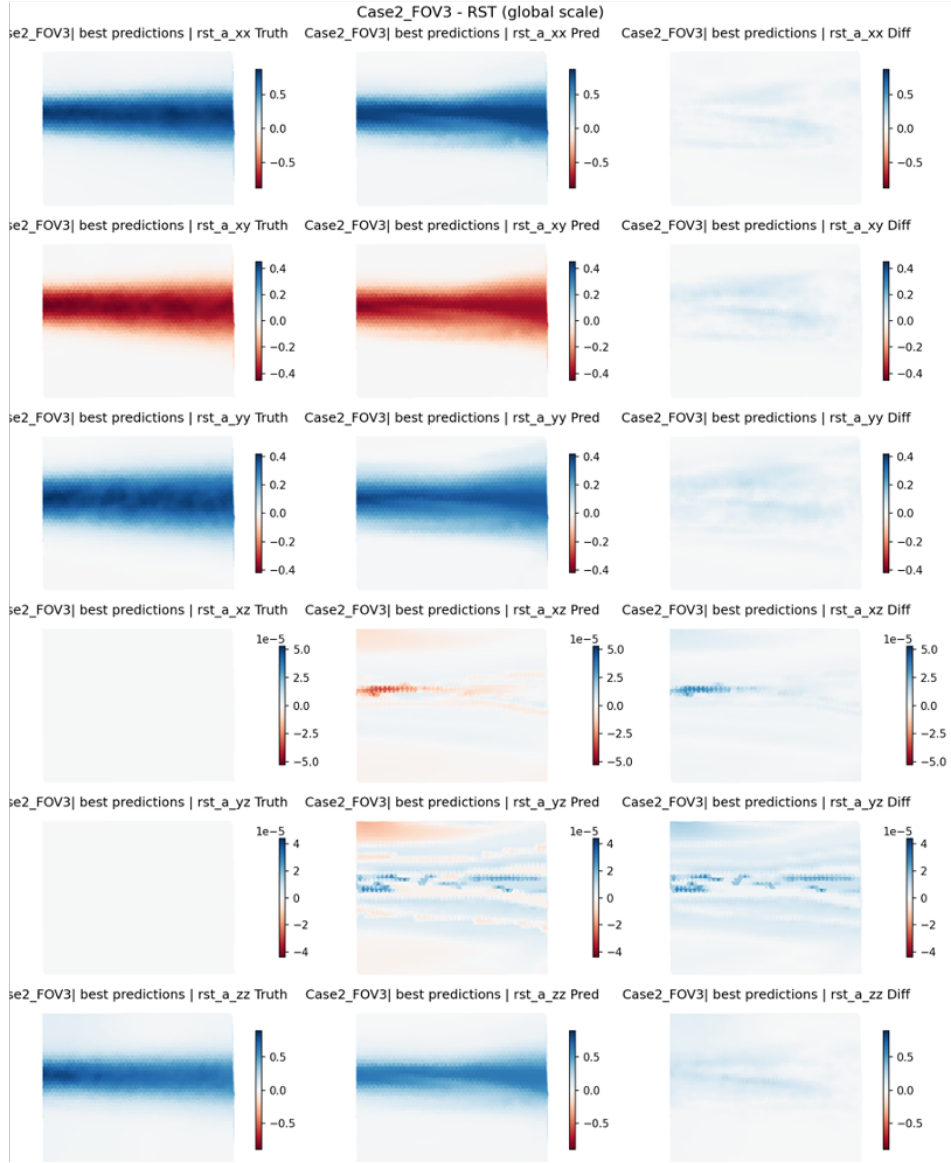


Figure 8: Predicted versus true Reynold's stress for the interpolation testing case.

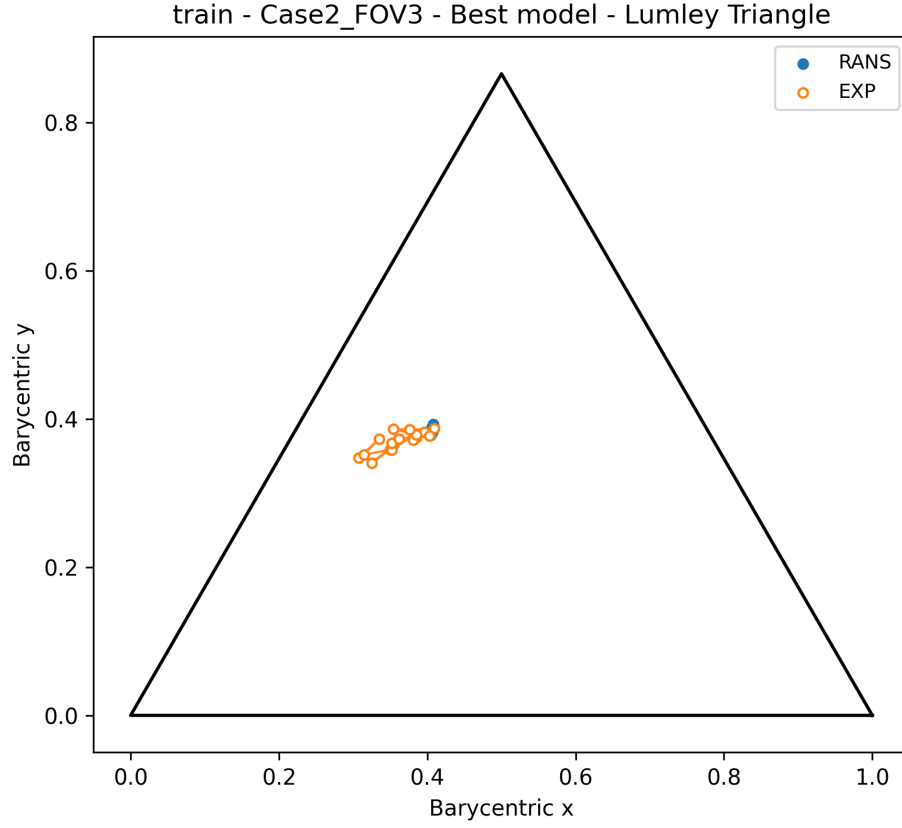


Figure 9: Barycentric Triangle of Reynolds stress anisotropy for the interpolation testing case.

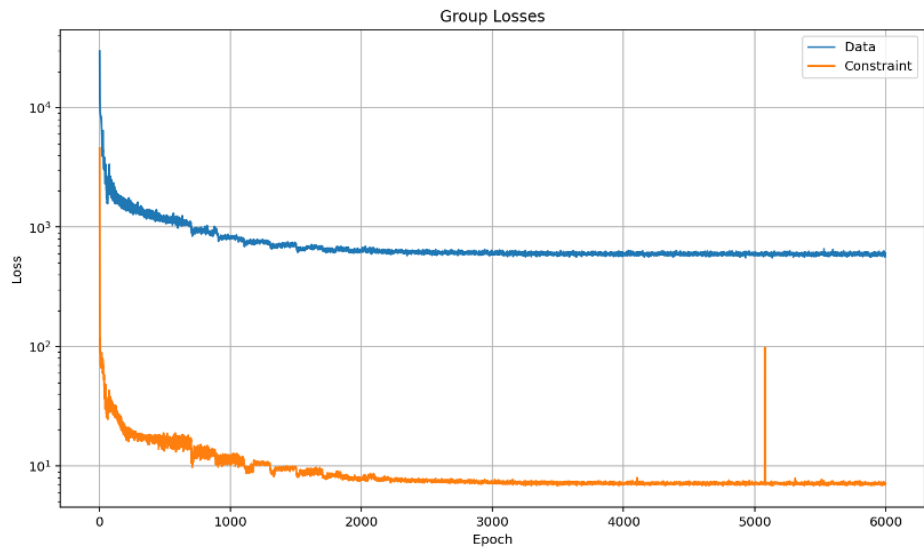


Figure 10: Loss history over the groups of data, physics and constraints for interpolation.

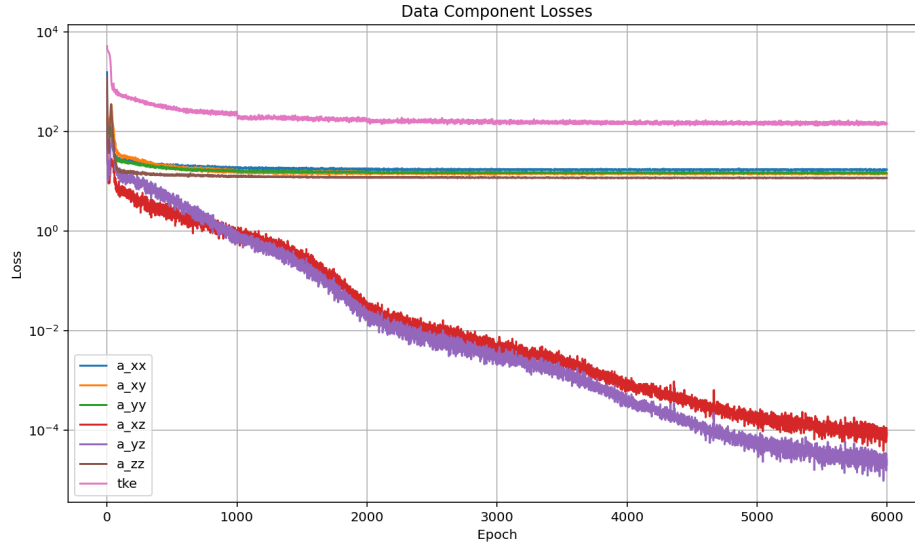


Figure 11: Loss history over the anisotropy and tke of data components for interpolation.

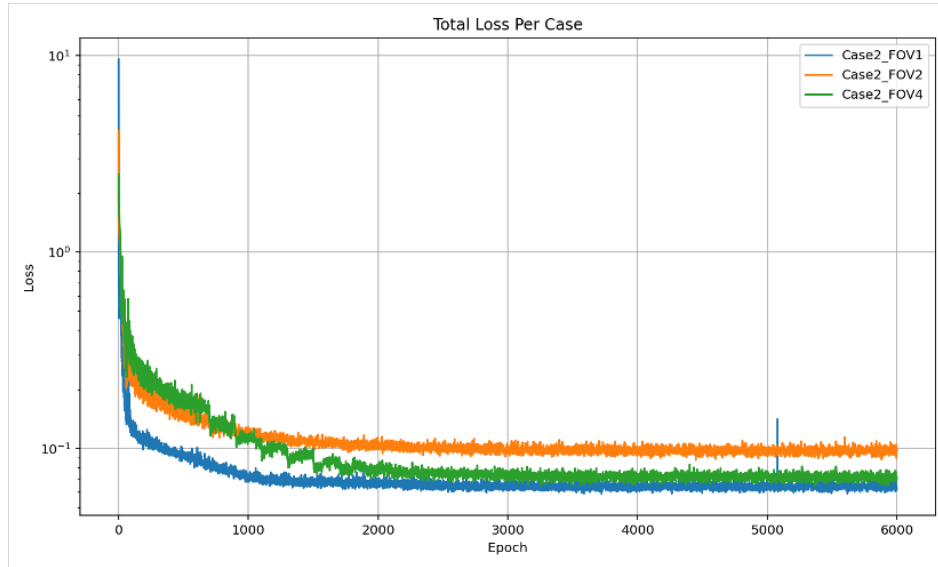


Figure 12: Loss history over each of the training cases for interpolation.

This trial displayed that the model was able to learn anisotropy and turbulent kinetic energy accurately. Since the model was provided all but the missing frame, it was able to interpolate within a turbulence regime whose growth and anisotropy trends were already bounded by training data. Over time, as invariant learning improved, the near-zero components ( $a_{xz}, a_{yz}$ ) also converged, and enforcing the zero-trace penalty further stabilized training. However, while the recovered stresses captured the overall shape, realizability was not enforced, as evidenced in the barycentric plots (Figure 9).

## 5.2 Single-Case Extrapolation Results

In this trial, the initial FOV was held out, requiring the model to extrapolate upstream of the training data. This is slightly tougher, as the model will need to perform well for the initial mixing region.

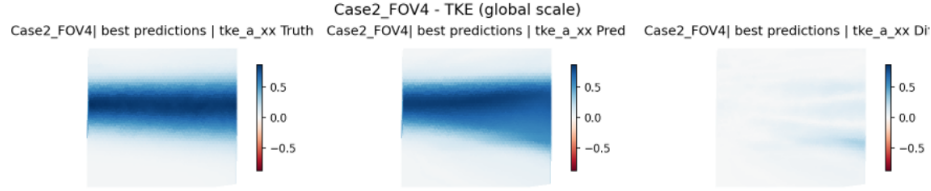


Figure 13: Predicted versus true turbulent kinetic energy (TKE) for the extrapolation case.

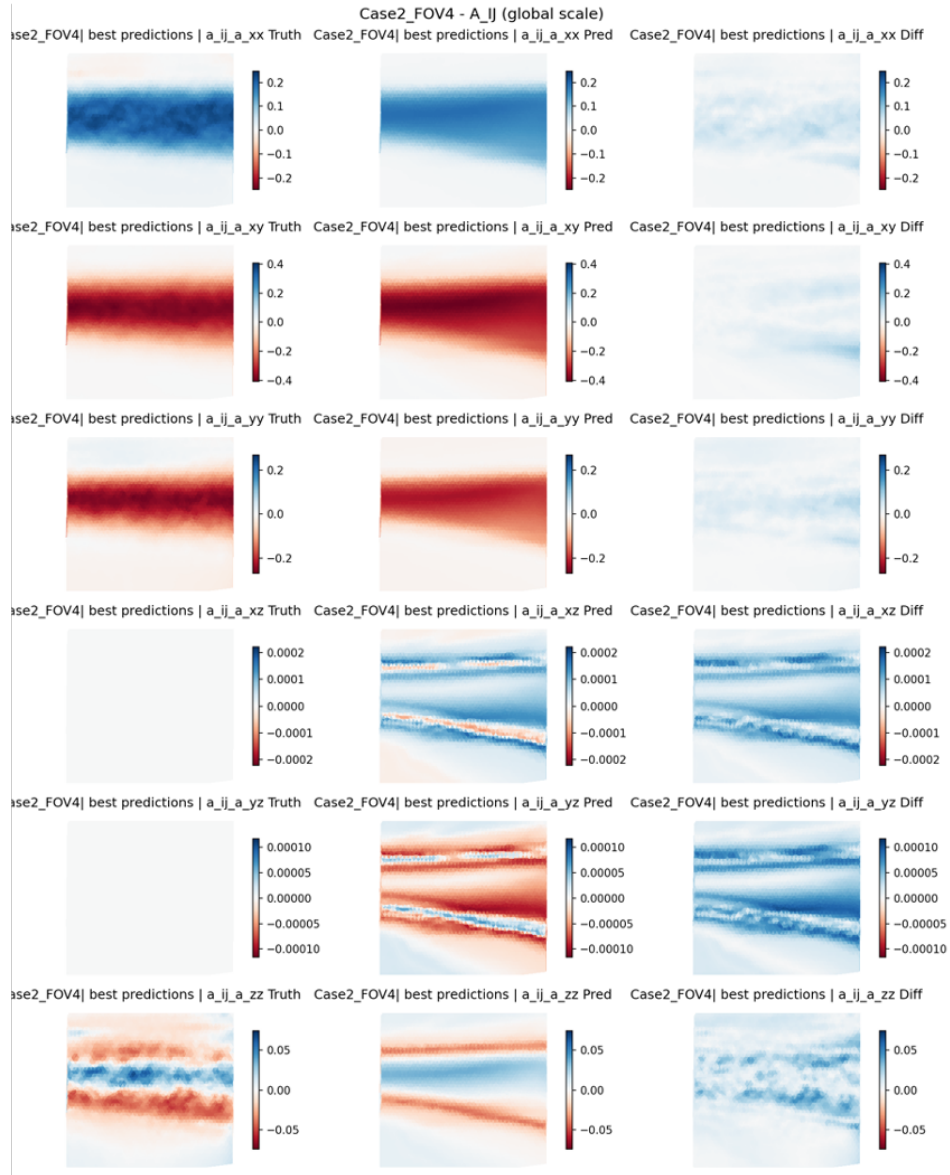


Figure 14: Predicted versus true anisotropy for the extrapolation case.

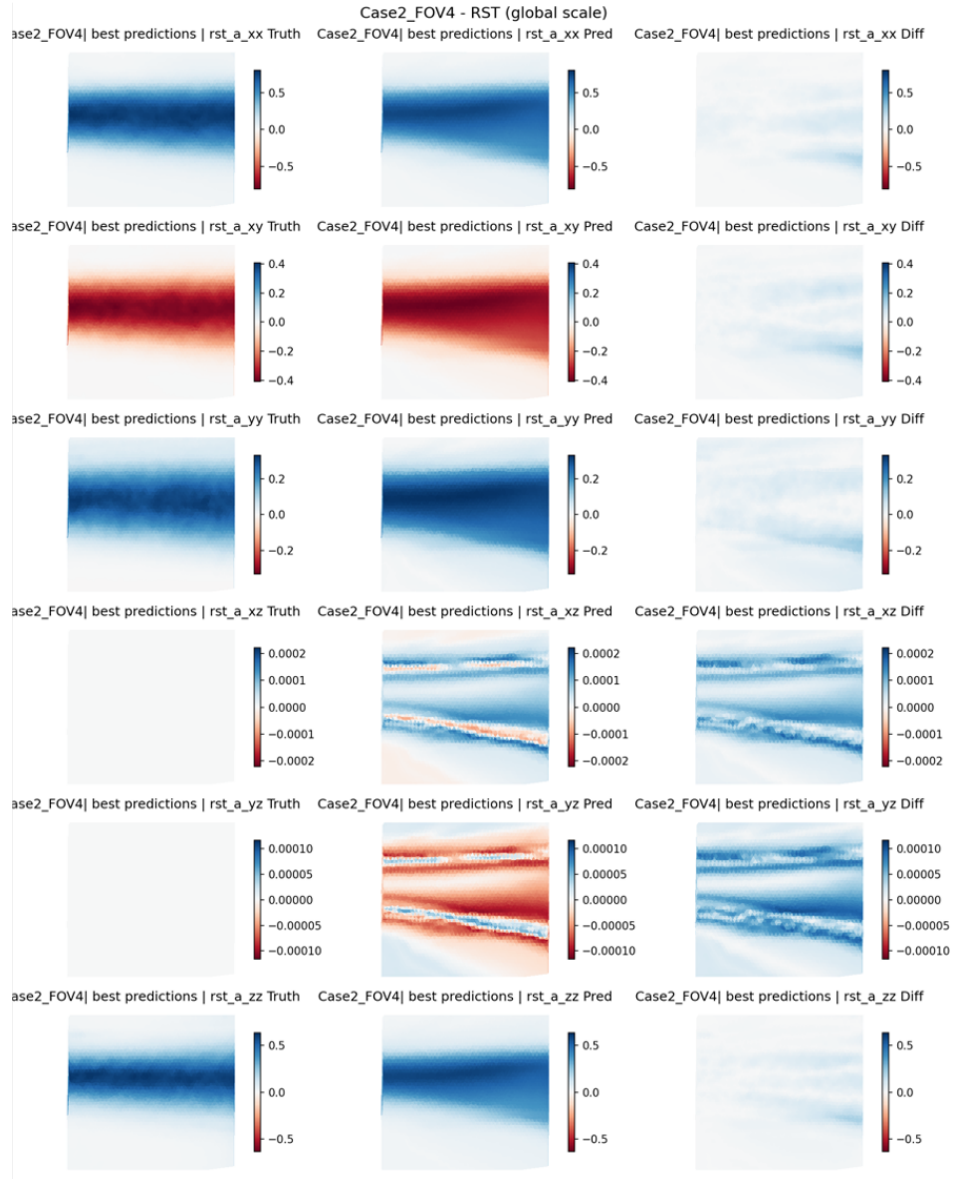


Figure 15: Predicted versus true Reynold's stress for the extrapolation case.

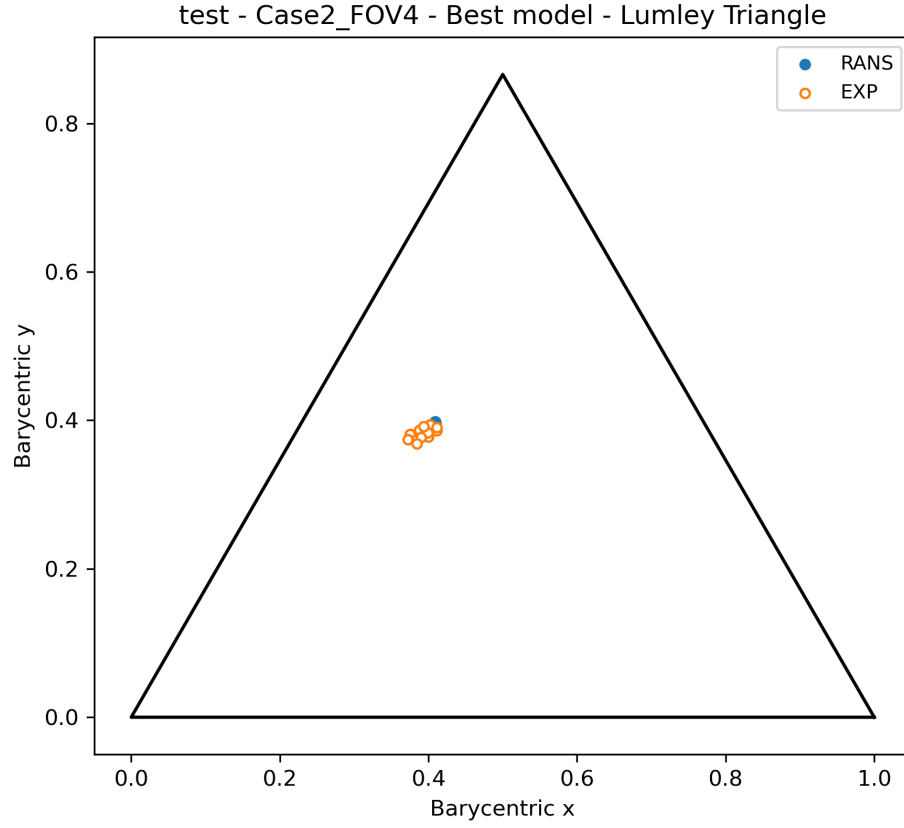


Figure 16: Barycentric Triangle of Reynolds stress anisotropy for the extrapolation testing case.

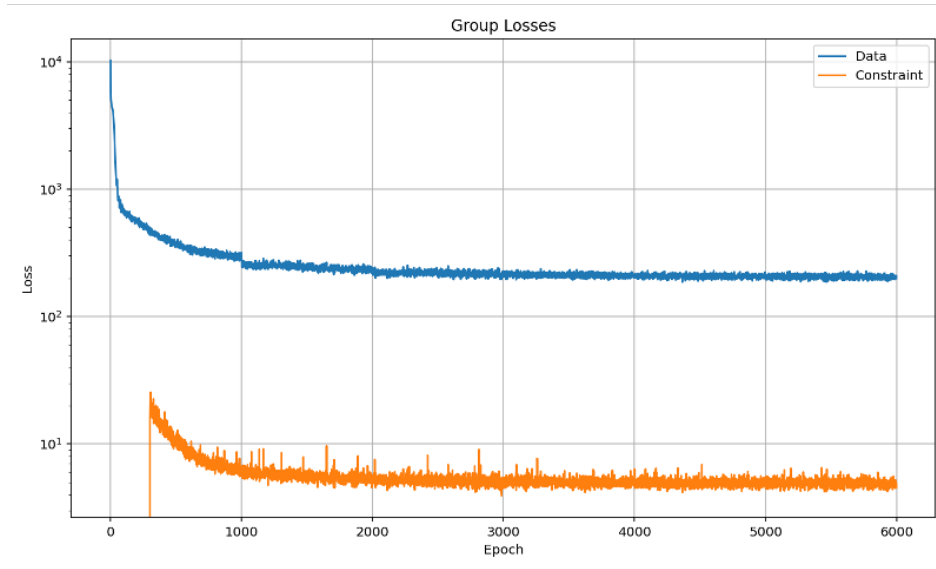


Figure 17: Loss history over the groups of data, physics and constraints for extrapolation.



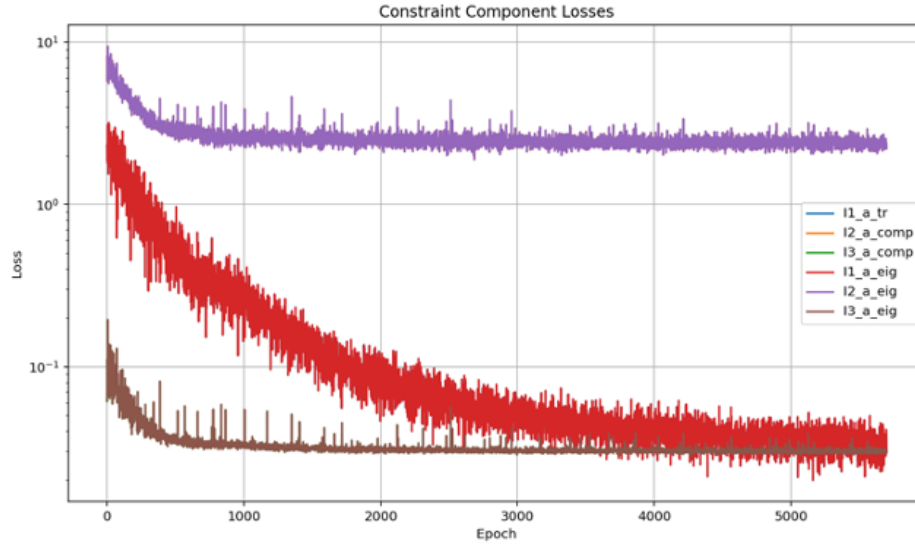


Figure 18: Loss history over the anisotropy and TKE of data components for extrapolation.

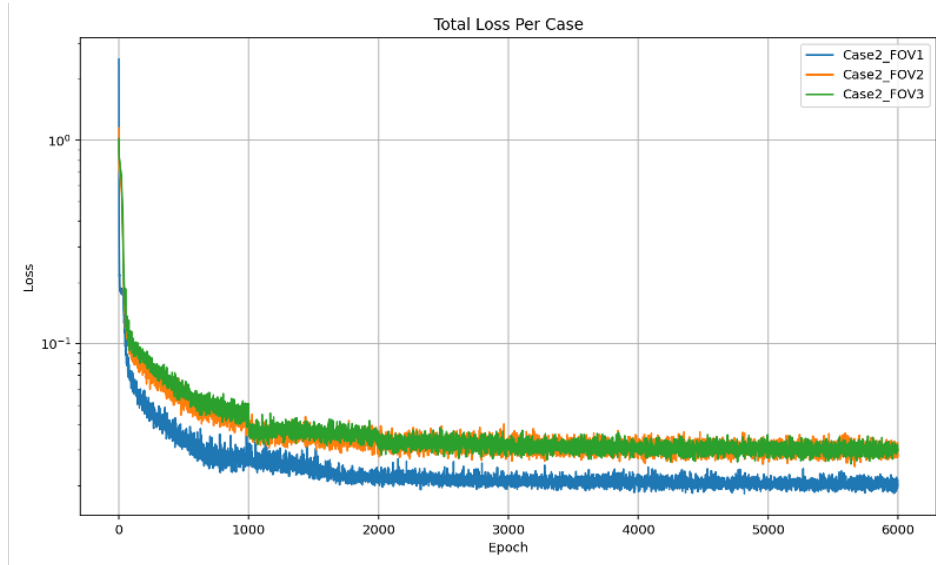


Figure 19: Loss history over each of the training cases for extrapolation.

For the extrapolation trial, the model displayed accuracy in recovering TKE and anisotropy. The Reynolds stress prediction, derived from the  $a_{ij}$  and TKE predictions, also displayed high levels of accuracy. It is worth noting that the scale of the  $xz$ ,  $yz$ , and  $zz$  terms is  $\mathcal{O}(10^{-5})$  relative to  $\mathcal{O}(10^0)$  for the dominant terms. This scaling difference suggests that small components are more prone to relative error amplification, making static component weights necessary to prevent them from being ignored by the optimizer. Loss curves indicated that the model was able to learn the data components but stagnated over time, underscoring the need for a more structured learning curriculum. As with interpolation, realizability was not strictly preserved, with predicted anisotropy drifting outside the realizable region of the Lumley triangle (Figure 16).

## 6 Discussion

The single-case trials demonstrated that interpolation was considerably easier than extrapolation. In the interpolation case, the model recovered TKE and anisotropy with high accuracy, while in the extrapolation case performance degraded in the upstream FOV. Even so, the general shape of the turbulence regime and its growth were clearly captured.

A key finding was that delaying the incorporation of constraint losses until several thousand epochs yielded the best overall results. As shown in Figure 10, this early stage “data only” training allowed the model to establish stable correlations between input features and Reynolds stresses before constraint enforcement began. This effectively served as a first step toward curriculum learning, which will be a necessary framework for future work.

Constraint losses themselves proved highly valuable. The zero-trace constraint on the anisotropy tensor was effective, ensuring realizable component balances, and the invariant losses provided additional structure by guiding the predicted anisotropy within the Lumley triangle. At the same time, extracting eigenvalues from predicted tensors introduced additional numerical sensitivity, since the implementation depends on PyTorch’s eigenvalue module. Future implementations may benefit from log-Euclidean metrics or alternative SPD-aware losses to reduce this sensitivity.

Overall, the model was able to correlate Favre-averaged, non-dimensional input features to turbulent kinetic energy and anisotropy. However, it was unable to preserve the mathematical structure of the tensors themselves. As a result, realizability and the Lumley anisotropy structure were not respected in the predictions. This finding reinforces that curriculum learning will be crucial in future work, ensuring that constraint enforcement is introduced in a staged manner that balances accuracy and physical consistency.

## 7 Next Steps and Future Work

### 7.1 Short Term

The immediate priority is to refine the learning curriculum. Training all loss terms simultaneously made convergence difficult. A staged strategy—first emphasizing TKE, then  $A_{ij}$  components, then invariants, and finally realizability (via Riemann distance or SPD constraints)—is expected to provide a more stable progression. Constraint losses should also be introduced with explicit scheduling rather than all at once.

### 7.2 Medium Term

Two key directions are envisioned:

- **Realizability Enforcement:** Explicitly constrain outputs to remain symmetric positive semi-definite (no negative or complex eigenvalues). This ensures predicted anisotropy lies within the realizable Lumley domain.

- **Closure Validation:** Compare residuals of the momentum equation, computed from input features, against gradients of predicted Reynolds stresses. This provides a physics-based check on whether the model approximates a closure.
- **Expanded Feature Set:** Beyond the Favre momentum features used here, new non-dimensional inputs can be explored: local convective Mach number  $M_c$ , local Reynolds number  $Re$ , local pressure coefficient  $C_p$ , and the  $Q$ -criterion (strain vs. rotation dominance). These features were not included in the present work but represent natural extensions.

### 7.3 Long Term

The long-term vision is to extend the framework to more complex flows. Applications to turbomachinery trailing edges will test generalizability in strongly inhomogeneous, geometry-driven turbulence. Once anisotropy prediction is robust, the framework can be expanded to include Favre energy-equation terms and turbulent heat fluxes, enabling joint modeling of turbulence and heat transfer.

## 8 Conclusion

### 8.1 Summary

This work demonstrated that recovery of turbulence anisotropy from RANS data is feasible, but not yet trivial. The current framework can successfully correlate Favre-averaged features to anisotropy and TKE, but realizability remains an unresolved challenge. Curriculum learning, progressing from the largest scales (TKE) to the smallest (invariants and constraints), emerged as the most promising path toward robust training. The methodology developed here can be refined to support either turbulence closure modeling or broader turbulence representation.

### 8.2 Significance

Achieving a closed-form anisotropy recovery solution is a significant step toward physics-informed data-driven turbulence modeling. In particular, applications to turbomachinery trailing edges hold strong promise, where improved prediction of anisotropy and Reynolds stresses could enhance blade design, cascade layout, and efficiency. Such a tool would provide engineers with a practical way to capture turbulence effects that are otherwise inaccessible in RANS. The primary caveat is the reliance on feature engineering pipelines, which require computation of gradients and derived terms—a process that currently adds 5–20 minutes per case.

### 8.3 Outlook

Future work should focus on enforcing realizability, refining curriculum learning strategies, and incorporating expanded non-dimensional features. These steps will bring the framework closer to providing a general-purpose, data-driven closure model for complex flows. Ultimately, extending the methodology to include heat transfer will pave

the way toward joint closure of the Favre momentum and energy equations, enabling direct application to real-world turbomachinery problems.

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## A Background

The instantaneous momentum equation from the compressible Navier–Stokes equations is:

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} \quad (24)$$

where:  $\rho$  is the density,  $u_i$  is the instantaneous velocity component,  $p$  is the pressure, and  $\tau_{ij}$  is the viscous stress tensor.

### A.1 Reynolds Averaging and the RANS Equations

For incompressible flows, the Reynolds decomposition of a velocity component  $u_i$  is:

$$u_i = \bar{u}_i + u'_i \quad (25)$$

where  $\bar{u}_i$  is the time-averaged velocity and  $u'_i$  is the fluctuating component.

Applying this decomposition to (24) and averaging yields the incompressible RANS momentum equation:

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial \overline{u'_i u'_j}}{\partial x_j} \quad (26)$$

The final term,  $\overline{u'_i u'_j}$ , is the Reynolds stress tensor, which introduces additional unknowns and requires a closure model.

### A.2 Favre Averaging and the FARANS Equations

For compressible flows, density-weighted (Favre) averaging is used:

$$u_i = \tilde{u}_i + u''_i \quad (27)$$

$$\tilde{u}_i = \frac{\overline{\rho u_i}}{\bar{\rho}} \quad (28)$$

where  $\tilde{u}_i$  is the Favre-averaged velocity and  $u''_i$  is the Favre fluctuation.

The Favre-averaged continuity equation is:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_j} (\bar{\rho} \tilde{u}_j) = 0 \quad (29)$$

The momentum equation in FARANS form becomes:

$$\frac{\partial}{\partial t} (\bar{\rho} \tilde{u}_i) + \frac{\partial}{\partial x_j} (\bar{\rho} \tilde{u}_i \tilde{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} - \frac{\partial \overline{\rho u''_i u''_j}}{\partial x_j} \quad (30)$$

Here, the last term  $\overline{\rho u''_i u''_j}$  is the Favre-averaged Reynolds stress tensor, which requires a closure model to solve.

### A.3 Reynolds Stress Tensor and Anisotropy

For compressible flow the Reynolds stress tensor is defined as:

$$R_{ij} = \overline{\rho u_i'' u_j''} \quad (31)$$

The turbulent kinetic energy (TKE) is:

$$k = \frac{1}{2} R_{ii} = \frac{1}{2} \text{tr}(R_{ij}) \quad (32)$$

The anisotropy tensor is:

$$a_{ij} = R_{ij} - \frac{2}{3} k \delta_{ij} \quad (33)$$

Then the normalized anisotropy tensor is:

$$b_{ij} = \frac{a_{ij}}{2k} - \frac{1}{3} \delta_{ij} \quad (34)$$

### A.4 Anisotropy Invariants and the Lumley Triangle

The invariants of  $b_{ij}$  are defined as:

$$II_b = b_{ij} b_{ji} \quad (35)$$

$$III_b = b_{ij} b_{jk} b_{ki} \quad (36)$$

Similarly, for  $a_{ij}$ , the invariants are:

$$I_a = a_{ii} = \text{tr}(a_{ij}) = 0 \quad (37)$$

$$II_a = a_{ij} a_{ji} \quad (38)$$

$$III_a = a_{ij} a_{jk} a_{ki} \quad (39)$$

If  $\lambda_1$ ,  $\lambda_2$ , and  $\lambda_3$  are the eigenvalues of  $b_{ij}$ , then invariants can also be expressed as:

$$I_b = \lambda_1 + \lambda_2 + \lambda_3 = 0 \quad (40)$$

$$II_b = -(\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_3 \lambda_1) \quad (41)$$

$$III_b = \lambda_1 \lambda_2 \lambda_3 \quad (42)$$

You can get the second and third invariants directly from the eigenvalues of  $b_{ij}$  using (41) and (42). In the  $(II_b, III_b)$  space, only certain combinations of these values are physically possible, and these make up the Lumley triangle. RANS models that use the Boussinesq hypothesis predict points that lie along a specific line in this triangle, called the “plane of strain.”

### A.5 Non-dimensional Forms

Let  $U_{\text{ref}}$  and  $L_{\text{ref}}$  be reference velocity and length scales. The non-dimensional variables are:

$$\hat{x}_i = \frac{x_i}{L_{\text{ref}}}, \quad \hat{u}_i = \frac{u_i}{U_{\text{ref}}}, \quad \hat{t} = \frac{t U_{\text{ref}}}{L_{\text{ref}}} \quad (43)$$

The non-dimensional FARANS equations follow from substituting these variables into (29) and (30).

## A.6 Lumley Triangle Coordinates from Eigenvalues

The barycentric map of turbulence states can be constructed directly from the ordered eigenvalues  $\lambda_1 \geq \lambda_2 \geq \lambda_3$  of the normalized anisotropy tensor  $b_{ij}$ . The three limiting turbulence states correspond to:

- **One-component turbulence (1C):**  $(\lambda_1, \lambda_2, \lambda_3) = (\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3})$ , representing all kinetic energy aligned in a single direction.
- **Two-component turbulence (2C):**  $(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3})$ , representing motion confined to a plane.
- **Isotropic turbulence (3C):**  $(0, 0, 0)$ , representing equal energy in all directions.

The barycentric coordinates  $(C_1, C_2, C_3)$  are defined as:

$$C_1 = \lambda_1 - \lambda_2, \quad C_2 = 2(\lambda_2 - \lambda_3), \quad C_3 = 3\lambda_3 + 1, \quad (44)$$

with  $C_1 + C_2 + C_3 = 1$ . These coefficients are then used to interpolate the location of the turbulence state within the Lumley triangle spanned by the 1C, 2C, and 3C limiting states.



## B Kurtosis-Based Norms

Feature-engineered gradient terms often exhibit heavy-tailed distributions with strong kurtosis. Direct min-max scaling can therefore be distorted by a small number of extreme outliers, forcing the bulk of the data into a narrow band near zero.

To address this, we applied a kurtosis-based clipping procedure before normalization. For each feature column, raw minima and maxima were clipped according to the expected spread implied by its kurtosis. This procedure ensured that extreme outliers did not dominate the scaling, while still mapping the vast majority of non-outlier values into the  $[0, 1]$  range.

The effect is to preserve the relative scale hierarchy across features while preventing numerical collapse of skewed distributions during training.

### Normalization Clipping



#### For features with large tails

- Clip the percentile from which to choose the maximum and minimum based on the tail direction

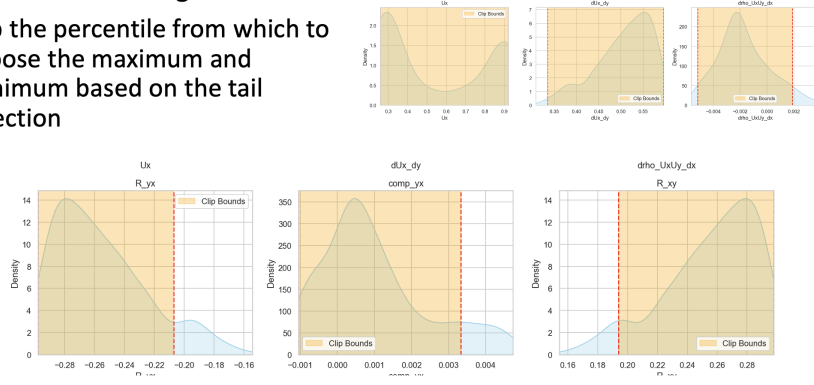


Figure 20: Visualization of Kurtosis Clipping

## C MLS Gradients

**Moving Least Squares (MLS) Gradients.** For improved gradient calculation and expression, we deployed a Moving Least Squares (MLS) approach. In this method, the algorithm loops through all points in the mesh and computes the gradient at each point with respect to all neighbors within a specified radius. For the shear mixing cases considered here, the radius was set to 10% of the reference length. A Gaussian kernel was used for weighting, so that the contribution of each neighbor to the local gradient decreases smoothly with its distance from the point under analysis. This formulation provides robust, geometry-aware gradients that outperform simple finite differences on irregular grids, particularly for capturing shear and rotational features in turbulent flows.